

AVOIDING FURSTENBERG KESTEN

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1. 11/10/10 *V*-Variability via Types

We show how *V*-variable trees can be constructed by using “types” instead of working with *V*-tuples of sets.

(We follow the usual notation and conventions for finite sequences involving truncation, concatenation, length and initial segment.)

In the following fix $V \in \mathbb{N}$ and fix a family $\mathbf{F}$ of IFSs.

Consider finitely branching trees T with a single root node and no terminating branches. In the applications each node will have at least two children. More precisely:

Definition 1.1. A *tree* T is a subset of the set of finite sequences of elements from $\mathbb{N}$.

- (1) If $\emptyset$ is the empty sequence then $\emptyset \in T$.
- (2) Suppose $\mathbf{i} \in T$. Then $\mathbf{ij} \in T$ iff $1 \leq j \leq N$, for some $N \geq 2$ depending on $\mathbf{i}$. We call N the *branching number* at $\mathbf{i}$.
- (3) If $\mathbf{i} \in T$ then $\mathbf{j} \in T$ for any $\mathbf{j} \preceq \mathbf{i}$.

Then $\emptyset$ is called the *root node*, the nodes $\mathbf{ij}$ in (2) are the *children* of $\mathbf{i}$ for $1 \leq j \leq N$, and the (unique) *parent* of $\mathbf{i}$ is $\mathbf{i}|(k-1)$ where $k = |\mathbf{i}|$ is the length of $\mathbf{i}$.

Definition 1.2. A *V*-variable tree T is a certain kind of *labelled tree*. Each node $\mathbf{i} \in T$ is assigned a pair $(\tau^{\mathbf{i}}, F^{\mathbf{i}})$ where $\tau^{\mathbf{i}} \in \{1, \dots, V\}$ is called the *type assigned to $\mathbf{i}$* , and $F^{\mathbf{i}} \in \mathbf{F}$ is called the *IFS assigned to $\mathbf{i}$* .

The branching number $N^{\mathbf{i}}$ at $\mathbf{i}$ is the cardinality of $F^{\mathbf{i}} = (f_1^{\mathbf{i}}, \dots, f_{N^{\mathbf{i}}}^{\mathbf{i}})$. The function $f_k^{\mathbf{i}}$ corresponds to the k th edge from $\mathbf{i}$, joining $\mathbf{i}$ to $\mathbf{ik}$.

If two nodes $\mathbf{i}$ and $\mathbf{j}$ at the same level have the same type they are assigned the same IFS, and their corresponding children are assigned corresponding types.

Summarising:

$$(1) \quad |\mathbf{i}| = |\mathbf{j}| \ \& \ \tau^{\mathbf{i}} = \tau^{\mathbf{j}} \implies \begin{cases} F^{\mathbf{i}} = F^{\mathbf{j}} \\ N^{\mathbf{i}} = N^{\mathbf{j}} =: N \\ \tau^{\mathbf{ik}} = \tau^{\mathbf{jk}} \end{cases} \quad \text{if } 1 \leq k \leq N$$

The *set of *V*-variable trees* is denoted by $\mathcal{T} = \mathcal{T}_V$.

In order to construct *V*-variable trees it is convenient to introduce the following notion.

Definition 1.3. We say $\tilde{F} = (F, \tau_1, \dots, \tau_N)$ is an *IFS/type vector* if $F \in \mathbf{F}$, N is the cardinality of F and $\tau_1, \dots, \tau_N \in \{1, \dots, V\}$. The *set of IFS/type vectors* is denoted by $\tilde{\mathbf{F}}$.

The informal idea in the following construction is as follows.

Initially one assigns a type to the root node, and w.l.o.g. the type is 1.

At stage (1) in the construction an IFS/type vector $\tilde{F}(v)$ is selected for each $v \in \{1, \dots, V\}$. Using $\tilde{F}(1)$ an IFS is assigned to the root node and a type to each of its children, the level 1 nodes.¹

At stage (2) an IFS/type vector $\tilde{F}(v)$ is selected for each $v \in \{1, \dots, V\}$. For each level 1 node, if v is the type of the node, then via $\tilde{F}(v)$ an IFS is assigned to the node and a type is assigned to each of its children. In particular a type is assigned to every level 2 node.

At stage (n) an IFS/type vector $\tilde{F}(v)$ is selected for each $v \in \{1, \dots, V\}$. For each level $n - 1$ node, if v is the type of the node, then via $\tilde{F}(v)$ an IFS is assigned to the node and a type is assigned to each of its children. In particular a type is assigned to every level n node.

Construction 1.4. A type $\tau^\emptyset \in \{1, \dots, V\}$ is assigned to the root node $\emptyset$. W.l.o.g. let $\tau^\emptyset = 1$. The tree is constructed by recursion as follows.

- (1) Select $\tilde{F}(v) \in \tilde{\mathbf{F}}$ for $v = 1, \dots, V$. (Note that the type 1 has already been assigned to the level 0 node.) Assign an IFS and a branching number to the level 0 node $\emptyset$, and a type to each level 1 node k , by

$$F^\emptyset = F, \quad N^\emptyset = N, \quad \tau^k = \tau_k, \quad \text{where } F(1) = (F, \tau_1, \dots, \tau_N) \text{ and } 1 \leq k \leq N.$$

- (2) Select $\tilde{F}(v) \in \tilde{\mathbf{F}}$ for $v = 1, \dots, V$. (Note that a type has already been assigned to each level 1 node.) Assign an IFS and a branching number to each level 1 node ℓ , and a type to each level 2 node ℓk , by setting

$$F^\ell = F, \quad N^\ell = N, \quad \tau^{\ell k} = \tau_k, \quad \text{where } F(\tau^\ell) = (F, \tau_1, \dots, \tau_N) \text{ and } 1 \leq k \leq N.$$

- (n) Select $\tilde{F}(v) \in \tilde{\mathbf{F}}$ for $v = 1, \dots, V$. (Note that a type has already been assigned to each level $n - 1$ node.) Assign an IFS and a branching number to each level $n - 1$ node, and a type to each level n node, by

$$F^i = F, \quad N^i = N, \quad \tau^{i\ell} = \tau_k \quad \text{where } F(\tau^i) = (F, \tau_1, \dots, \tau_N) \text{ and } 1 \leq k \leq N.$$

Notation 1.5. The particular tree T will usually be understood from context.

We use the convention that quantities defined from the IFS or the type at the node $\mathbf{i} \in T$ are denoted with *superscript* $\mathbf{i}$. For example, the type, the IFS and the branching number at $\mathbf{i}$ are denoted by $\tau^{\mathbf{i}}$, $F^{\mathbf{i}}$ and $N^{\mathbf{i}}$ respectively. Similarly we write $F^{\mathbf{i}} = (f_1^{\mathbf{i}}, \dots, f_{N^{\mathbf{i}}}^{\mathbf{i}})$.

We associate the function $f_j^{\mathbf{i}}$ to the j th *edge* with initial node $\mathbf{i}$ and hence with terminal node $\mathbf{i}j$. Similarly, let $\ell_j^{\mathbf{i}}$ be the Lipschitz constant of $f_j^{\mathbf{i}}$.

Other quantities associated to the node $\mathbf{i}$ will depend not only on $(\tau^{\mathbf{i}}, F^{\mathbf{i}})$ but will be obtained by multiplying or composing certain quantities or functions along the sequence of edges connecting $\emptyset$ to $\mathbf{i}$. We will usually use a *subscript* $\mathbf{i}$ in this case. In particular, we multiply the Lipschitz constants and compose the contraction maps associated with the edges joining $\emptyset$ to $\mathbf{i}$ to obtain the following with $k = |\mathbf{i}|$:

$$\begin{aligned} f_{\mathbf{i}} &= f_{i_1}^\emptyset \circ f_{i_2}^{i_1} \circ f_{i_3}^{i_1 i_2} \circ \dots \circ f_{i_k}^{i_1 i_2 \dots i_{k-1}}, \\ \ell_{\mathbf{i}} &= \ell_{i_1}^\emptyset \cdot \ell_{i_2}^{i_1} \cdot \ell_{i_3}^{i_1 i_2} \cdot \dots \cdot \ell_{i_k}^{i_1 i_2 \dots i_{k-1}}. \end{aligned}$$

2. The Probability Distribution on the Set of V -Variable Trees

Fix a probability distribution P on $\mathbf{F}$.

Definition 2.1. The probability distribution on $\tilde{\mathbf{F}}$ is the distribution of the random vector $\tilde{F} = (F, v_1, \dots, v_N)$, where F is a random IFS with distribution P , N is the cardinality of F and $v_1, v_2, \dots$ are iid random variables uniformly distributed on $\{1, \dots, V\}$ independent of F .

We denote this probability distribution on $\tilde{\mathbf{F}}$ also by P .

¹The $\tilde{F}(v)$ selected at stage 1 for $v \neq 1$ are not used in the construction, and are included only for consistency with the construction at later stages.

The V -variable tree T in Construction 1.4 is uniquely determined by the sequence

$$\mathcal{E}_1, \mathcal{E}_2, \dots, \mathcal{E}_n, \dots,$$

where $\mathcal{E}_n$ is the V -tuple of IFS/type vectors selected at level n . This leads to the following definition.

Definition 2.2. The *distribution* $P = P_V$ on the set $\mathcal{T}_V$ of V -variable trees is obtained by selecting each of the V IFS/type vectors at stage n in Construction 1.4 according to the distribution P on $\tilde{\mathbf{F}}$, in an iid manner and independent of the IFS/type vectors at all other stages.

Notation 2.3. Realisations of a random V -variable tree may be denoted by T^ω , but usually the ω is suppressed.

In situations where more than one tree is being considered, an appropriate superscript will again be used, with parentheses if needed for clarity. For example, if T is a given tree and $\mathbf{i} \in T$ then the subtree rooted at $\mathbf{i}$ is denoted $T^{\mathbf{i}} = T^{(\mathbf{i})}$. In particular, $T^{(\emptyset)} = T$. If $\mathbf{j}$ is a node of $T^{(\mathbf{i})}$ then the corresponding node of T is $\mathbf{ij}$. Similarly, $F^{(\mathbf{i})\cdot\mathbf{j}} = F^{\mathbf{ij}} = F^{(\emptyset)\cdot\mathbf{ij}}$; that is, the IFS at node $\mathbf{j}$ of the subtree $T^{(\mathbf{i})}$ is the same as the IFS at node $\mathbf{ij}$ of the tree T , and the tree T is just the tree $T^{(\emptyset)}$. Similar notation applies to other quantities besides F .

If $n(1) < n(2) < \dots$ is the sequence of necks for T , then $T^{\mathbf{i}} = T^{\mathbf{j}}$ if $|\mathbf{i}| = |\mathbf{j}| = n(k)$ for some k . In this case we define $T^{(k)} = T^{\mathbf{i}}$ and $F^{(k)\cdot\mathbf{j}} = F^{(\mathbf{i})\cdot\mathbf{j}} = F^{\mathbf{ij}}$ for any $\mathbf{i}$ with $|\mathbf{i}| = n(k)$. Again, similar notation applies to other quantities besides F .

3. Avoiding Furstenberg Kesten

Assume

$$(2) \quad \begin{aligned} N_{\max} &:= \sup \{N : F = (f_1, \dots, f_N) \in \mathbf{F}\} < \infty \\ \ell_{\max} &:= \sup \{ \text{Lip } f_j : F = (f_1, \dots, f_N) \in \mathbf{F}, 1 \leq j \leq N \} < 1. \end{aligned}$$

Recall that the sequence of necks follows a geometric distribution. Let q be the probability of a neck occurring at any particular level.

Assume the V -variable random tree T has distribution P_V . Let $T = T^\omega$ be a realisation.

For $\mathbf{i} \in T$ and $|\mathbf{i}| = n$ define

$$r_{\mathbf{i}} = r_{i_1}^\emptyset \cdot r_{i_2}^{i_1} \cdot r_{i_3}^{i_1 i_2} \cdot \dots \cdot r_{i_n}^{i_1 \dots i_{n-1}},$$

where $r_k^{\mathbf{i}} = (\ell_k^{\mathbf{i}})^\alpha = (\text{Lip } f_k^{\mathbf{i}})^\alpha$ and $\alpha > 0$. Let $r_{\max} = \ell_{\max}^\alpha$.

Theorem 3.1. *The expected exponential growth rate of $\sum_{|\mathbf{i}|=n} r_{\mathbf{i}}$, namely*

$$(3) \quad \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E} \log \sum_{|\mathbf{i}|=n} r_{\mathbf{i}} =: \gamma =: \gamma(\alpha),$$

exists and is finite. Moreover,

$$(4) \quad \lim_{n \rightarrow \infty} \frac{1}{n} \log \sum_{|\mathbf{i}|=n} r_{\mathbf{i}} = \gamma \text{ a.s.}$$

Also, one has the formula

$$(5) \quad \gamma = q \mathbb{E} \log \sum_{|\mathbf{i}|=n(1)} r_{\mathbf{i}},$$

Proof. We begin by considering limits through the sequence of necks:

$$(6) \quad \lim_{k \rightarrow \infty} \frac{1}{n(k)} \log \sum_{|\mathbf{i}|=n(k)} r_{\mathbf{i}}.$$

Then

$$\begin{aligned} & \sum_{\mathbf{i} \in T, |\mathbf{i}|=n(k)} r_{\mathbf{i}} \\ &= \sum_{\mathbf{i} \in T, |\mathbf{i}|=n(k)} r_{i_1}^\emptyset \cdot r_{i_2}^{i_1} \cdot r_{i_3}^{i_1 i_2} \cdot \dots \cdot r_{i_{n(k)}}^{\mathbf{i}(n(k)-1)} \end{aligned}$$

$$\begin{aligned}
&= \sum_{\mathbf{i} \in T, |\mathbf{i}|=n(k)} \left\{ \left(r_{i_1}^{\emptyset} \cdot r_{i_2}^{i_1} \cdot r_{i_3}^{i_1 i_2} \cdot \dots \cdot r_{i_{n(1)}}^{\mathbf{i}|(n(1)-1)} \right) \right. \\
&\quad \cdot \left(r_{i_{n(1)+1}}^{\mathbf{i}|n(1)} \cdot r_{i_{n(1)+2}}^{\mathbf{i}|n(1)+1} \cdot \dots \cdot r_{i_{n(2)}}^{\mathbf{i}|(n(2)-1)} \right) \cdot \dots \\
&\quad \cdot \left. \left(r_{i_{n(k-1)+1}}^{\mathbf{i}|n(k-1)} \cdot r_{i_{n(k-1)+2}}^{\mathbf{i}|n(k-1)+1} \cdot \dots \cdot r_{i_{n(k)}}^{\mathbf{i}|(n(k)-1)} \right) \right\} \\
&= \left(\sum_{\mathbf{i} \in T^{(0)}, |\mathbf{i}|=n(1)} r_{i_1}^{(0), \emptyset} \cdot r_{i_2}^{(0), i_1} \cdot r_{i_3}^{(0), i_1 i_2} \cdot \dots \cdot r_{i_{n(1)}}^{(0), \mathbf{i}|(n(1)-1)} \right) \\
&\quad \cdot \left(\sum_{\mathbf{i} \in T^{(1)}, |\mathbf{i}|=n(2)-n(1)} r_{i_1}^{(1), \emptyset} \cdot r_{i_2}^{(1), i_1} \cdot r_{i_3}^{(1), i_1 i_2} \cdot \dots \cdot r_{i_{n(2)-n(1)}}^{(1), \mathbf{i}|(n(2)-n(1)-1)} \right) \cdot \dots \\
&\quad \cdot \left(\sum_{\mathbf{i} \in T^{(k-1)}, |\mathbf{i}|=n(k)-n(k-1)} r_{i_1}^{(k-1), \emptyset} \cdot r_{i_2}^{(k-1), i_1} \cdot r_{i_3}^{(k-1), i_1 i_2} \cdot \dots \cdot r_{i_{n(k)-n(k-1)}}^{(k-1), \mathbf{i}|(n(k)-n(k-1)-1)} \right) \\
(7) \quad &= \left(\sum_{|\mathbf{i}|=n(1)} r_{\mathbf{i}}^{(0)} \right) \cdot \left(\sum_{|\mathbf{i}|=n(2)-n(1)} r_{\mathbf{i}}^{(1)} \right) \cdot \dots \cdot \left(\sum_{|\mathbf{i}|=n(k)-n(k-1)} r_{\mathbf{i}}^{(k-1)} \right)
\end{aligned}$$

The first and last equality are immediate from the definitions. The second equality is just a bracketing of terms.

For the third equality note that each $n(j)$ is a neck and recall Notation 2.3. A term such as $r_{i_{n(1)+1}}^{\mathbf{i}|n(1)}$, which corresponds to the edge in T from $\mathbf{i}|n(1) = i_1 \dots i_{n(j)}$ to $i_1 \dots i_{n(1)} i_{n(1)+1}$, is independent of $\mathbf{i}|n(1)$ and can also be regarded as corresponding to the level one edge from $\emptyset$ to $i_{n(1)+1}$ of the unique tree $T^{(1)}$ rooted at every level $n(1)$ node. Thus we rewrite $r_{i_{n(1)+1}}^{\mathbf{i}|n(1)}$ as $r_{i_1}^{(1), \emptyset}$, with an abuse of notation in that $\mathbf{i}$ and $i_{n(1)+1}$ in the first term refer to words from $T = T^{(0)}$ whereas i_1 in the second term is the first element of a word from $T^{(1)}$. Similarly, $r_{i_{n(1)+2}}^{\mathbf{i}|n(1)+1}$ is also independent of $\mathbf{i}|n(1)$ and can also be regarded as corresponding to a level two edge from $T^{(1)}$, etc. Now use simple algebra analogous to

$$\sum_{a \in A, b \in B, c \in C} abc = \left(\sum_{a \in A} a \right) \left(\sum_{b \in B} b \right) \left(\sum_{c \in C} c \right).$$

to put the summations inside the parentheses.

Let

$$X_k = \log \sum_{|\mathbf{i}|=n(k)-n(k-1)} r_{\mathbf{i}}^{(k-1)}, \quad k \geq 1.$$

By construction the X_k are iid and in particular

$$X_1 = \log \sum_{|\mathbf{i}|=n(1)} r_{\mathbf{i}}.$$

We compute

$$\begin{aligned}
\mathbb{E} |X_1| &\leq \sum_{n \geq 1} P(n(1) = n) \max \left\{ |\log (N_{\max}^n r_{\max}^n)|, |\log (N_{\min}^n r_{\min}^n)| \right\} \\
&= \max \left\{ |\log (N_{\max} r_{\max})|, |\log (N_{\min} r_{\min})| \right\} \mathbb{E} n(1) \\
&= q^{-1} \max \left\{ |\log (N_{\max} r_{\max})|, |\log (N_{\min} r_{\min})| \right\}.
\end{aligned}$$

Similarly,

$$\mathbb{E}(X_1)^2 \leq \max \left\{ \log^2 (N_{\max} r_{\max}), \log^2 (N_{\min} r_{\min}) \right\} \mathbb{E} n^2(1) < \infty.$$

It follows from the strong law of large numbers that the limit in (6), with $n(k)$ replaced by k , exists a.s. and equals $\mathbb{E} X_1$. But $k/n(k) \rightarrow q$ as $k \rightarrow \infty$ by a standard property of geometric distributions. This establishes the theorem for sequences of necks.

For arbitrary n let n^* be the unique neck equal to, or immediately preceding, n . It follows as in (7) that the error term in replacing n by n^* in the expression $\frac{1}{n} \log \sum_{|i|=n} r_i$ in (4) is dominated by $(n - n^*)(1/n) \log(N_{\max} r_{\max}) \rightarrow 0$ a.s.

The theorem now follows. $\square$

Remark 3.2.

- (1) One can show for $\alpha \geq 0$ that $\gamma(\alpha)$ is Lipschitz, monotonically decreasing, with upper and lower bounds on the Lipschitz constants. Since $\gamma(0) > 0$ there is a unique α such that $\gamma(\alpha) = 0$.

One can then show that this α gives the a.s. dimension of the associated fractal, assuming the open set condition.

- (2) One can take other quantities besides ℓ , such as crossing times, effective resistance, etc. provided the appropriate independencies still hold.

$\square$

4. OLD: 7/9/09: Avoiding Furstenberg Kesten

The key idea is due to Bob Scealy, see [Sce08, Chapter 5.1] and [Sce07]. The following is essentially self contained, with notation that of [BHS].

4.1. Notation. Consider a V -variable V -tuple $(K_1, \dots, K_V)$ of fractals with address

$$(8) \quad a_1 a_2 a_3 \dots$$

Let $M^i = M^{a_i} = M^i(\alpha)$ be the corresponding flow matrices as in [BHS, Definition 4.2].

Let $S_n = S_n(\alpha)$ be the sum of the diameters to the power α of the components of the level n prefractal approximation to K^1 . See [BHS, (4.4)] where a slightly different notation is used.

Using the notation of [BHS], with open set $\mathcal{O}$ for the open set condition, rescale and define

$$(9) \quad r_{\emptyset}(\omega) := |\overline{\mathcal{O}}| = 1, \quad r_{m_1 \dots m_n}(\omega) := |\overline{\mathcal{O}}_{m_1 \dots m_n}| = r_{m_1}^{\omega(\emptyset)} \cdot \dots \cdot r_{m_n}^{\omega(m_1 \dots m_{n-1})} \text{ for } m_1 \dots m_n \in \omega,$$

where ω is the construction tree for K_1 . Then, suppressing ω ,

$$(10) \quad S_n(\alpha) := \sum_{\{\mathbf{m} \in \omega: |\mathbf{m}|=n\}} |\overline{\mathcal{O}}_{\mathbf{m}}|^\alpha = \sum_{\{\mathbf{m} \in \omega: |\mathbf{m}|=n\}} (r_{\mathbf{m}}(\omega))^\alpha, \text{ noting } S(\omega, 0, \alpha) = 1.$$

Moreover, S_n is the sum of the entries in the first row of $M^1 \circ \dots \circ M^n$, i.e.

$$(11) \quad S_n = e^T M^1 \circ \dots \circ M^n \mathbf{1},$$

where e is the V -dimensional unit column vector with 1 as *first* entry, the transpose e^T is the corresponding row vector, and $\mathbf{1}$ is the V -dimensional column vector of units. See [BHS, Proposition 4.3].

For purposes here, (11) can be taken as the definition of S_n .

4.2. Overview. We are interested in showing the existence of an a.s. fixed growth rate $d = d(\alpha)$. That is,

$$\log S_n(\alpha) \sim nd(\alpha) \text{ a.s. as } n \rightarrow \infty.$$

Equivalently,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log S_n(\alpha) = d(\alpha) \text{ a.s., } \left(\text{and so } d(\alpha) = \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E} \log S_n(\alpha) \right).$$

In fact, the Furstenberg Kesten theorem shows this is true if we replace S_n , the sum of the elements in the first row of $M^1 \circ \dots \circ M^n$, by the sum of all elements. A further argument then shows the same result follows for any row, and in particular the first. See [BHS].

It is then not difficult to show $d(\alpha)$ is decreasing in α and has a unique zero. It is plausible, and is proved in [BHS], that this gives the a.s. Hausdorff dimension of K_1 , or indeed of any K_v .

The Furstenberg Kesten theorem can be avoided by using (17) to see that

$$S_n = e^T M^1 \circ \dots \circ M^n \mathbf{1} = \lambda_1 \dots \lambda_k e^T E \mathbf{1},$$

and then arguing as in Theorem 4.9. The point is that λ_i is an eigenvalue with left eigenvector as in (17). Alternatively, one can see this directly from (10).

This gives, using the notation of Definition 4.3,

$$(12) \quad d(\alpha) = q \mathbb{E} \log \lambda(\alpha),$$

where $\lambda(\alpha) := S_\gamma(\alpha)$ and the random integer γ is the first neck.

In particular, the Hausdorff dimension is the unique α such that

$$(13) \quad \mathbb{E} \log \lambda(\alpha) (= \mathbb{E} \log S_\gamma(\alpha)) = 0.$$

4.3. Details.

Definition 4.1. One says p , or equivalently M^p , is a *neck* if all entries in M^p other than those in the first column are zero.

This is a little different from Definition 5.3 in [BHS], where M^p is a neck iff there is some column, not necessarily the first, such that all elements other than those in this column are zero.

Remark 4.2. If M^p is a neck then for any $V \times V$ matrix M , all elements in $M \circ M^p$ other than those in the first column are also zero.

It follows that e^T is a left eigenvector of $M \circ M^p$, i.e.

$$(14) \quad e^T M \circ M^p = \lambda e^T$$

for some λ . □

Definition 4.3. Choose an iid sequence $M^1, \dots, M^\gamma$ until the first neck γ is obtained.

This defines an integer valued random variable γ . From (14) there is a real valued random variable λ defined by

$$(15) \quad e^T M^1 \circ \dots \circ M^\gamma = \lambda e^T.$$

Remark 4.4. From (11) and (15),

$$(16) \quad S_\gamma(\alpha) = e^T M^1 \circ \dots \circ M^\gamma \mathbf{1} = \lambda(\alpha),$$

and these are then the top left element of $M^1 \circ \dots \circ M^\gamma$.

Proposition 4.5. Let $1 \leq n_2 < n_3 < \dots$ be the necks in the sequence (8).

Let

$$(17) \quad \begin{aligned} e^T M^1 \circ \dots \circ M^{n_1} &= \lambda_1 e^T \\ e^T M^{n_{i-1}+1} \circ \dots \circ M^{n_i} &= \lambda_i e^T \text{ if } 2 \leq i \leq k, \end{aligned}$$

where n_k is the largest neck such that $n_k \leq n$.

Let

$$(18) \quad E = \begin{cases} M^{n_k+1} \circ \dots \circ M^n & \text{if } n_k < n, \\ I & \text{if } n_k = n, \end{cases}$$

where I is the $V \times V$ identity matrix.

Then

$$(19) \quad S_n = \lambda_1 \cdot \dots \cdot \lambda_k e^T E \mathbf{1}.$$

Proof. Immediate from (11), (17) and (18). □

Remark 4.6. With the notation from (17), for $i \geq 1$,

$$(20) \quad \begin{aligned} \text{dist } \lambda &= \text{dist } \lambda_i, \\ \text{dist } \gamma &= \text{dist}(n_i - n_{i-1}), \end{aligned}$$

where we set

$$(21) \quad n_0 = 0.$$

Remark 4.7. Suppose q is the probability that M^i is a neck. Then γ is given by the geometric distribution

$$(22) \quad P(\gamma = j) = (1 - q)^{j-1} q$$

for $j \geq 1$. Moreover²

$$(23) \quad \mathbb{E}(\gamma) = q^{-1}.$$

²This is standard.

Remark 4.8. Assume a uniform upper bound on the number of functions in each IFS and a uniform lower bound on the greatest of the contraction ratios associated with each IFS.

It follows for some $0 < a < \infty$ depending on α that

$$(24) \quad a^{-p} \leq \sum_{i=1}^V (M^1 \circ \dots \circ M^P)_{1i} \leq a^p$$

for *any* sequence $M^1, \dots, M^P$ of allowable flow matrices.

Weaker conditions than (24) clearly suffice for the following theorem to hold. $\square$

Theorem 4.9. *Assuming (24),*

$$(25) \quad \lim_{n \rightarrow \infty} \frac{1}{n} \log S_n(\alpha) = q \mathbb{E} \log \lambda(\alpha) \quad (= q \mathbb{E} \log S_\gamma(\alpha)) \quad \text{a.s.},$$

where q and λ are as in Definition 4.3.

The Hausdorff dimension of K_1 is the unique α such that $\mathbb{E} \log \lambda(\alpha) \quad (= \mathbb{E} \log S_\gamma(\alpha)) = 0$.

Proof. Let n_k be the largest neck such that $n_k \leq n$ and note $k = k(n)$ is independent of α .

Then

$$\begin{aligned} \lim_{n \rightarrow \infty} \log S_n &= \lim_{n \rightarrow \infty} \frac{1}{n} \log(\lambda_1 \cdots \lambda_k e^T E \mathbf{1}) \quad \text{from (19)} \\ &= \lim_{n \rightarrow \infty} \frac{\sum_{i=1}^k \log \lambda_i + \log(e^T E \mathbf{1})}{\sum_{i=1}^k (n_i - n_{i-1}) + (n - n_k)} \\ &= \lim_{n \rightarrow \infty} \frac{\frac{1}{k} \sum_{i=1}^k \log \lambda_i + \frac{1}{k} \log(e^T E \mathbf{1})}{\frac{1}{k} \sum_{i=1}^k (n_i - n_{i-1}) + \frac{1}{k} (n - n_k)} \\ &= q \mathbb{E}(\log \lambda) \quad \text{a.e.} \end{aligned}$$

To see the last equality note

$$\frac{1}{k} \sum_{i=1}^k \log \lambda_i \rightarrow \mathbb{E} \log \lambda \quad \text{a.s.},$$

since the λ_i are iid with $\text{dist } \lambda_i = \text{dist } \lambda$ by (20), and by using the strong law of large numbers. Similarly

$$\frac{1}{k} \sum_{i=1}^k (n_i - n_{i-1}) \rightarrow \mathbb{E} \gamma = q^{-1},$$

since $n_i - n_{i-1}$ are iid with $\text{dist}(n_i - n_{i-1}) = \text{dist } \gamma$ by (20), and again by using the strong law of large numbers and (24). Moreover,

$$\frac{1}{k} (n - n_k) \leq \frac{1}{k} (n_{k+1} - n_k) \rightarrow 0 \quad \text{a.s.},$$

To see it let p be the probability of a neck and let γ be the first neck. Then

$$P(\gamma = k) = (1 - p)^{k-1} p.$$

and

$$\mathbb{E} \gamma = \sum_{k=1}^{\infty} k (1 - p)^{k-1} p.$$

Let

$$\phi(p) = \sum_{k=1}^{\infty} (1 - p)^{k-1} p = \frac{p}{1 - (1 - p)} = 1,$$

as is clear from the probabilistic interpretation of ϕ .

Hence

$$\begin{aligned} 0 = \phi'(p) &= \frac{d}{dp} \left(p + \sum_{k \geq 2} (1 - p)^{k-1} p \right) \\ &= 1 + \sum_{k \geq 2} (1 - p)^{k-1} - \sum_{k \geq 2} (k - 1) (1 - p)^{k-2} p \\ &= \sum_{k \geq 1} (1 - p)^{k-1} - \sum_{k \geq 1} k (1 - p)^{k-1} p \\ &= -\mathbb{E}(\gamma) + p^{-1}. \end{aligned}$$

since if $a_k := n_{k+1} - n_k$ then

$$\frac{a_k}{k} = \frac{a_1 + \dots + a_k}{k} - \frac{k-1}{k} \frac{a_1 + \dots + a_{k-1}}{k-1} \rightarrow q^{-1} - q^{-1} = 0 \quad \text{a.s.}$$

Finally, from (18) and (24),

$$\frac{1}{k} |\log(e^T E \mathbf{1})| \leq \frac{1}{k} (n - n_k) \log a \rightarrow 0 \quad \text{a.s.}$$

The last claim now follows from [BHS]. $\square$

5. 20/4/10: Construction of V -Variable Trees

Fix a family $\mathbf{F}$ of IFSs as before. For initial purposes it is sufficient only that the IFSs consist of uniformly contractive maps on $\mathbb{R}^d$.

Fix a natural number V .

Definition 5.1. (See Figure 1) An *IFS tree* T , corresponding to $\mathbf{F}$ and V , is a tree with the following properties:

- (1) there is a single, level 0, root node $\emptyset$;
- (2) the *branching number* $N^{\mathbf{i}} \geq 2$ at each node $\mathbf{i}$ is finite and ≥ 2 ;
- (3) the edges with initial node $\mathbf{i}$ are numbered (“left to right”) from $1, \dots, n^{\mathbf{i}}$; we write $\mathbf{i} = i_1 \dots i_k$ in the usual manner where $|\mathbf{i}| := k \geq 1$ is the *level* of $\mathbf{i}$;
- (4) there is an *IFS* $F^{\mathbf{i}} \in \mathbf{F}$ associated with each node $\mathbf{i}$, $N^{\mathbf{i}} = |F^{\mathbf{i}}|$ (the cardinality of $F^{\mathbf{i}}$), and the k th edge with initial node $\mathbf{i}$ is associated with the k th function in the IFS $F^{\mathbf{i}}$.

The unique compact set $K = K(T)$ associated with T in the usual manner is called a *recursive fractal*.

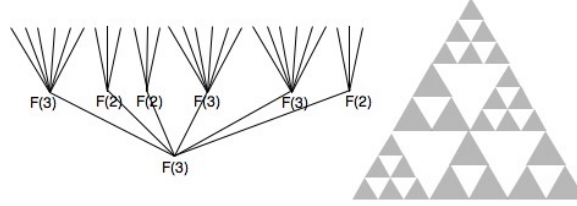


FIGURE 1. Level 2 approximations to an IFS tree and to the associated fractal. Here $\mathbf{F} = \{F(2), F(3)\}$ contains the IFSs generating $SG(2)$ and $SG(3)$ respectively. Edges of the tree with a given initial node are enumerated from left to right; they correspond to subcells enumerated anticlockwise from the bottom left corner of the cell corresponding to the given node.

Definition 5.2. (See Figure 2) A V -variable (IFS) tree is an IFS tree T with a *type* $\tau^{\mathbf{i}} \in \{1, \dots, V\}$ associated to each node $\mathbf{i}$, such that if two nodes at the *same* level k have the same type then:

- (1) both nodes have the same branching number and associated IFS, i.e.

$$|\mathbf{i}| = |\mathbf{j}| \ \& \ \tau^{\mathbf{i}} = \tau^{\mathbf{j}} \implies N^{\mathbf{i}} = N^{\mathbf{j}} \ \& \ F^{\mathbf{i}} = F^{\mathbf{j}};$$

- (2) comparable successor nodes have the same type as each other, i.e.

$$|\mathbf{i}| = |\mathbf{j}| \ \& \ \tau^{\mathbf{i}} = \tau^{\mathbf{j}} \ \& \ 1 \leq p \leq N^{\mathbf{i}} \implies \tau^{i^p} = \tau^{j^p}.$$

The recursive fractal $K = K(T)$ corresponding to the V -variable tree T is called a V -variable fractal. The *class of V -variable trees* and *class of V -variable fractals* corresponding to $\mathbf{F}$ are denoted by $\mathcal{T}_V^{\mathbf{F}} = \mathcal{T}_V$ and $\mathcal{K}_V^{\mathbf{F}} = \mathcal{K}_V$ respectively.

Remark 5.3. A 1-variable tree is essentially the same as an IFS tree which generates a scale irregular, i.e. homogeneous, fractal.

A V -variable tree has at most V distinct IFSs associated to the nodes at each fixed level. If two nodes at the *same* level of a V -variable tree have the same type then the subtrees rooted at these two nodes are identical, i.e.

$$|\mathbf{i}| = |\mathbf{j}| \ \& \ \tau^{\mathbf{i}} = \tau^{\mathbf{j}} \implies \sigma_{\mathbf{i}} T = \sigma_{\mathbf{j}} T.$$

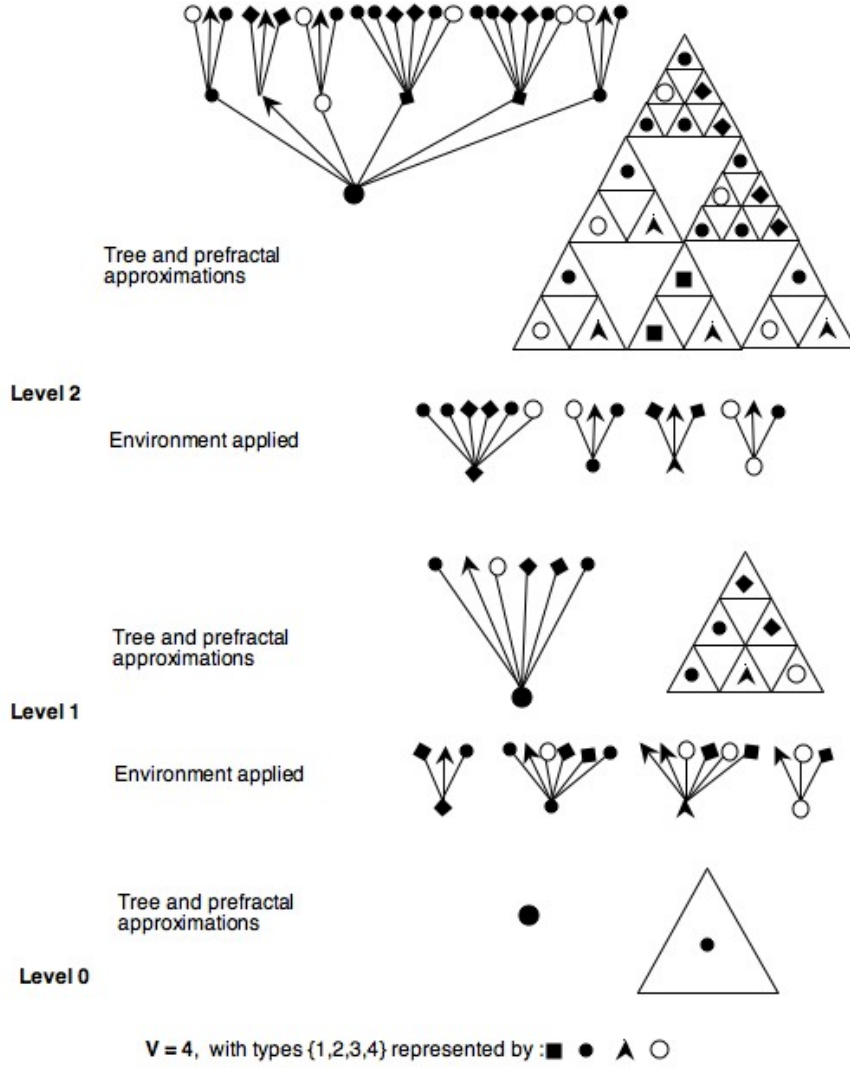


FIGURE 2. Approximations to a 4-variable tree and the prefractal approximations to the corresponding 4-variable fractal. The IFSs are $F(2)$ and $F(3)$ generating $SG(2)$ and $SG(3)$. The environment at each level is applied to the approximation at the previous level. The IFS labels are not shown but are here determined by the branching number.

In particular there are at most V distinct subtrees rooted at the *same* level. $\square$

The following is used in the construction and analysis of V -variable fractals.

Definition 5.4. An *environment* $\mathcal{E}$ assigns to each type $v \in \{1, \dots, V\}$ an IFS $F = F_v \in \mathbf{F}$ and a sequence of types $(\tau_{v1}, \dots, \tau_{vN_v})$, where $N_v = |F_v|$. We write

$$\mathcal{E} = (E_1^\mathcal{E}, \dots, E_V^\mathcal{E}), \quad E_v^\mathcal{E} = (F_v^\mathcal{E}, \tau_{v1}^\mathcal{E}, \dots, \tau_{vN_v}^\mathcal{E}).$$

For the following consider the case $n = 2$ in Figure 2.

Construction 5.5. V -variable trees are constructed recursively as follows:

- (1) *Stage 0:* Assign a type $\tau^\emptyset$ to the root node $\emptyset$.
- (2) Assume that by the completion of stages 0 to $n - 1$:
 - (a) the tree and all nodes $\mathbf{i}$ with $|\mathbf{i}| \leq n - 1$ have been constructed,
 - (b) a type $\tau^{\mathbf{i}}$ has been assigned if $|\mathbf{i}| \leq n - 1$,
 - (c) an IFS $F^{\mathbf{i}}$ has been assigned if $|\mathbf{i}| \leq n - 2$,
 - (d) the two properties in Definition 5.2 hold if $|\mathbf{i}| \leq n - 2$.

- (3) *Stage n* : Choose an environment $\mathcal{E}$. We follow the notation of Definition 5.4 and use $\mathcal{E}$ in the obvious way to assign an IFS to the level $n - 1$ nodes, to construct the level n nodes, and to assign a type to every level n node.

More precisely, for each node $\mathbf{i}$ with $|\mathbf{i}| = n - 1$, if $v = \tau^{\mathbf{i}}$ is the type already assigned to $\mathbf{i}$, then:

- (a) $F^{\mathbf{i}} := F_v^{\mathcal{E}}$ is the IFS assigned to $\mathbf{i}$,
- (b) the branching number $N^{\mathbf{i}}$ at $\mathbf{i}$ is $|F^{\mathbf{i}}|$,
- (c) for $1 \leq j \leq |F^{\mathbf{i}}|$ the type $\tau^{\mathbf{i}j} := \tau_{vj}^{\mathcal{E}}$ is the type assigned to the level n node $\mathbf{i}j$.

It follows by an easy induction that the properties in Definition 5.2 hold at all nodes. $\square$

Definition 5.6. The environment $\mathcal{E}$ in Definition 5.4 is a *neck* if all $\tau_{vi}^{\mathcal{E}}$ are equal. A *neck occurs at level n* in the construction of a V -variable tree if the environment $\mathcal{E}$ applied at stage n is a neck environment.

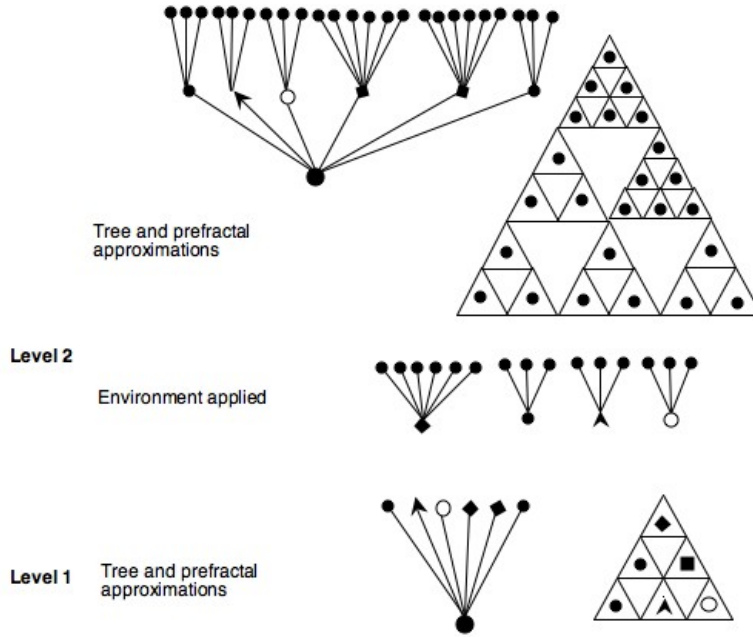


FIGURE 3. Compare with Figure 2, except that now a neck occurs at level 2. All subtrees rooted at this level will be the same, although they have not yet been constructed. All 2-complexes will be the same up to scaling by factors determined by the construction up to this level.

If a neck occurs at level n then the type assigned to every node at that level is the same. See Figure 3. It follows from Remark 5.3 that all subtrees rooted at level n will be the same. Note that the subtrees are only constructed at later stages, and even the common value of the IFS at each level n node is not determined until stage $n + 1$.

There is however no restriction on the IFSs occurring in a neck environment $\mathcal{E}$. For a level n neck these IFSs are applied at level $n - 1$.

Because there is an upper bound on the number of functions N^F in any IFS $F \in \mathbf{F}$, there is only a finite number of type choices to be made in selecting an environment. It follows that necks occur infinitely often almost surely with respect to the probability P_V defined in Definition 5.8. The sequence of neck levels in the construction of a V -variable tree or neck is denoted by

$$0 = n(0) < n(1) < \dots < n(i) < \dots$$

Definition 5.7. Fix a probability distribution P on $\mathbf{F}$. This induces a *probability distribution P_V on the set of environments* as follows. Choose the IFSs $F_v^{\mathcal{E}}$ for $v \in \{1, \dots, V\}$ in an iid manner according to P . Choose types $\tau_{vj}^{\mathcal{E}}$ for $1 \leq j \leq |F_v^{\mathcal{E}}|$ in an iid manner according to the uniform distribution on $\{1, \dots, V\}$ and otherwise independently of the $F_v^{\mathcal{E}}$.

Definition 5.8. The *probability distribution on the set $\mathcal{T}_V$* of V -variable trees is obtained by choosing τ^0 according to the uniform distribution and independently choosing the environments at each stage in an iid manner according to P_V . This probability distribution on V -variable trees induces a *probability distribution on the set $\mathcal{K}_V$* of V -variable fractals. Both the distributions on trees and on fractals are also denoted by P_V .

The σ -algebra of measurable subfamilies of $\mathcal{K}_V$ is the Borel σ -algebra induced by the Hausdorff metric on nonempty compact subsets of $\mathbb{R}^d$. For $\mathcal{T}_V$ one uses any of the standard tree metrics.

Random V -variable trees and *random V -variable fractals* are random labelled trees and random compact subsets of $\mathbb{R}^d$ respectively with the distribution P_V . We write T or T^ω , and K or K^ω , respectively.

It will normally be convenient to identify the sample space for our random quantities (fractals, trees, etc) with the set of V -variable IFS trees. For example, $\sigma_i \omega$ is just the transfer operator defined in Notation ?? for a tree T .

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